RobTek Master’s Thesis guidelines

2024-30-09 – Version 1.2

This document describes the RobTek (Robot Systems Engineering at SDU) education committee recommendations on technical guidelines for MSc theses submitted on the RobTek MSc civil engineering program at SDU.

The intent of these guidelines is to give students an equal basis for documenting their project and give supervisors and censors a common reference for forming the evaluation.

It is emphasized that these are recommendations, and that supervisors are free to lay down other guidelines or specify guidelines, within the first month of the project, if they so whish.

This document is structured as follows:

Mandatory prerequisites:

The Master's thesis is completed on the last year of the master programme. The work must follow the semester-structure as all other courses.

30 ECTS Master’s Thesis:

Before entering into an agreement on submission of the thesis (the thesis contract), the student must have obtained at least 85 ECTS during his/her master programme, including all mandatory course subjects.

40 ECTS Master’s Thesis:

Before entering into an agreement on submission of the thesis (the thesis contract), the student must have obtained at least 60 ECTS during his/her master programme, including all mandatory course subjects on the 1st and 2nd semester.

To make a 40 ECTS Master’s Thesis the thesis topic must contain a significant experimental part.

Overall Timeline:

40 ECTS		
Spring semester / Summer		
	During Spring semester	Find a company or a topic to work with
	20 -30 May	Register for the thesis (as if it was a course)
	May/June	Contact potential supervisors
	31 August	Last date for Project contract
The following Fall semester		
	1 September	Start working on the thesis
The following Spring semester		
	First workday in June	Submission of Thesis
	June	Exam

30 ECTS		
Fall semester		
	During Fall semester	Find a company or a topic to work with
	20 -30 November	Register for the thesis (as if it was a course)
	November/December	Contact potential supervisors
	31 January	Last date for Project contract
The following Spring semester		
	1 February	Start working on the thesis
	First workday in June	Submission of Thesis
	June	Exam

How to find a topic:

Master theses can be done in cooperation with companies, but you do not have to have a company included.

For finding a topic without a company the following channels are good ideas:

- Talk to your lecturers about opportunities
- Come to our event talking more about Master theses and the courses offered on the 3rd semester of the Master (usually in May)

If you want to find a company related project these venues might work:

- Participate in Internship and Project Day at The Faculty of Engineering, SDU and get inspired
- Look at SDU's jobbank (jobbank.sdu.dk)
- Look at https://www.odenserobotics.dk/member-directory/ for a list of interesting companies

What makes a good topic?

- Builds on something you know how to do
- Include something you do not know how to solve
- Demonstrate your good engineering and scientific skills:
    - Use a systematic approach to the problem and choices
    - Define clear goals and evaluate the quality of the result
    - Reflects on advantages and downsides of an approach
    - Demonstrate understanding of "state-of-the-art" and knowledge about all parts of the project
    - A negative result can be OK as long as good process is followed
- Should be the final piece of the study and combine everything that you have learned during the study

Supervisor:

You need to have a supervisor (from TEK) for your master thesis. (Typically, only Professors, Associate Professors, Assistant Professors, and in special cases External lecturers and Post docs are allowed to be main supervisors.) It is your responsibility to find a supervisor.

Contact a supervisor who you think matches your areas of interest or think can be a good supervisor for you.

Contact them via email or talk with them during class.

Work out the project outline with the supervisor. Talk to you supervisor about:

- Project content including the title and description
- Agree on the expectation for supervision based on a proposal from the supervisor
- How the collaboration with the company is organized including agreements on confidentiality. (Only relevant if a company is involved)
- In the group settle expectations for team roles and collaboration in the group. (Only relevant if you are doing your master thesis together with another student.)

Project contract:

The Project contract must be submitted digitally via SPOC. It must feature:

- Title that captures the area, problem, and solution
- Administrative info on the project
- Project description:
    - Text (1-2 pages) that describes the project including requirements
        - How is the project delimited?
        - What needs to be learned?
        - What theories and technologies are in play?
    - A good project description requires a good collaboration with your supervisor (and company)
    - A good project description takes time to develop
- Example on structure:
    - Context / problem / solution proposal / overall timeline

Be aware:

- It is not easy to change the title once the Project contract is submitted.
  If during the thesis work, you learn that the thesis takes you in a different direction than initially assumed you can make amendments in the contract. Please fill in the form as follows:
  https://spoc.sdu.dk -> Study Secretariat -> Other forms -> Amendment of title or Project Description - Final Project Contract, Bachelor Project Contract or Master's Thesis Contract.
  The changes must be approved by your supervisor. Once this is done, you will be notified by an e-mail. You must have submitted the application re. change of title or project description no later than one month before the submission deadline.
  As a main rule, the approval of change of title or research question does not qualify for dispensation as to submission deadline.
- The description may change
- Talk with your supervisor before you deviate from your plan
- A 40 ECTS master thesis needs to contain a significant experimental part. The project description needs to make that significant experimental part explicit.

If you are unsure about how to do this one of the resources below might help:

> *Dahl, Anders, Dich, Trine, Hansen, Tina & Olsen, Vagn (2009): Styrk projektarbejdet – en redskabsbog til problemorienteret projektarbejde, kap. 5.*
>
> *Christian W. Dawson: Projects in Computing and Information Systems. 2nd edition. Chapter 3 “Choosing a project and writing a proposal”*

Non-Disclosure Agreement:

If a company asks for an agreement on confidentiality (Non-Disclosure Agreement (NDA)) the student should contact SDU Rio via this form https://contracts.sdu.dk/. We do not recommend that the student signs any NDA before having contacted SDU Rio. If the NDA is not carefully crafted it can create significant problems in the student's future career and potentially contain parts that do not allow the university (supervisor / censor) to assess the work (and thereby you cannot get your degree).

Individual supervisors cannot sign NDAs. If the company requires an NDA from the supervisors, the University (as the supervisor's employer) will sign such an NDA. This needs to be taken up early and go through SDU Rio (see above).

The thesis report:

To attend the exam the students must hand-in a thesis report. The thesis project must include an abstract in English. If the report is in English, the abstract can be in Danish. The abstract is included in the assessment.

Length:

The number of pages should be regarded as maximum number of pages, but it is only recommendations. In general, it is recommended to be as clear and brief as possible while still being comprehensive and fully documenting the work. A normal page (NP) is here defined as 2400 characters including whitespace and counting all table and figure captions.

The page count should include the main text which is all text starting from the introduction and ending with the conclusion, i.e., excluding front page, abstract, preamble, preface... etc., table of contents and references and appendices.

Thesis or course projects: 1 NP pr Course ECTS plus 5 NP pr student in the group

Example: 30 ECTS thesis, group of two students: 30+2x5 = 40 NP

Frontpage:

The frontpage should as a minimum include:

- Type of project (e.g., Master thesis)
- Title of the project
- Names of students and their study number
- "University of Southern Denmark" and name of center, department or faculty
- Month and year
- Name(s) of supervisor(s)
- Character count
- Maybe a nice picture, sketch or diagram from your project

A latex thesis template, that you can use if wanted, can be found at: https://gitlab.sdu.dk/robtek-educations/thesis-template

Potential suggestion for report structure:

Please discuss your report structure with your supervisor.

The following can be seen as a template for a report structure. Don't necessarily follow it, but rather think about which parts fit your work and which don't and then adapt as wanted.

- Front page
  (See above)
- Title page
  At least:
  - Project title
  - Project period
  - Your name, email address etc.
  - Report type (e.g., Master thesis)
  - Your education (e.g., Robotics)
  - "University of Southern Denmark" and name of centre, department or faculty.
  - Names of supervisor(s)
  - Copyright and license statement for report and material in the electronic archive.
- Abstract
  Maximum one page
  Consider using 4 paragraphs following this template (without writing the titles)
  - Introduction
  - Materials and methods
  - Results and discussion
  - Conclusion and future work
- Abstract in Danish
- Reading guide
  Maximum one page
- List of abbreviations
- Foreword
- Personal foreword
- Table of content
  Consider using numbers for chapters and sections, if you are using Latex, set the depth to e.g., 3 or 4.
- Introduction
  - Background information
    Helicopter perspective, what is the project all about? Give the outside reader at chance to understand the domain.
  - Problem statement
    Wherein lies the problem? What will you try to solve?
  - Related work
    During your literature search what did you find?
    Your own or the research group's prior work if any?
  - Project aim or Hypothesis
    How did you derive your project aim (engineering design process) or hypothesis which you are testing (scientific method) and how it connects to related work?
- Main part 1
  - Meat and potato part, typically really 2-4 chapters dividing your work into some logical structure.
  - Consider using a common template for sections in all the meat and potato parts such as:
    - Introduction
    - Materials and methods / Background
    - Results

        - Discussion
        - Conclusion
- …
- Main part n
- Discussion
  Summing up all the main parts 1 to n conclusions.
- Conclusion
  Maximum one page
    - In general, focusing a lot on the discussion and concluding if the project aims or hypothesis was met partly or fully.
    - Do not write anything that is not written elsewhere in the report.
    - Remember to describe future work at the end of the conclusion.
- List of figures
- List of tables
- List of literature / References
- Appendix 1
- …
- Appendix n

Consider beginning each chapter with 2-5 lines describing what this particular chapter covers. Make sure that this is well aligned with your reading guide.

Hand In:

The thesis report must be submitted electronically in Digital Eksamen, https://digitaleksamen.sdu.dk/. The report must be submitted as one single pdf document. Any enclosures must be submitted as separate files. The submission deadline is stated in the thesis contract. Normally it is the first weekday of June.

If you have been granted dispensation for a later deadline this deadline will be stated in the letter of dispensation.

If you do not submit the thesis report within the deadline, you will use an examination attempt and you must sign an agreement supplementary to the thesis contract within two weeks of the submission deadline. You find the supplementary contract through https://spoc.sdu.dk -> Study Secretariat -> Master thesis supervisor agreements. You must enclose an extended research question with the supplementary agreement. The extended research question must be within the original subject area, and it must, regardless of the number of ECTS awarded, equal a workload of an additional three months. You can get a third attempt according to the same rules as apply to the second attempt. The deadline will be extended by three months.

If you fail the master thesis exam the same rules applies as if you did not meet the deadline (see above).

Oral Examination:

The examination time is typically 60 minutes per student. Typically, the following format is used for the exam:

For 1 (2) student(s):

1. 5 (10) minutes evaluation of thesis (without students)
2. 15 (30) Presentation by the students of the main goals and results and if relevant a demonstration of a robot prototype to visualize the end product.
3. 30 minutes individual examination
4. 10 (20) minutes evaluation and feedback

Additional information:

- Other relevant information can be found on our webpage for master theses: https://mitsdu.dk/en/mit_studie/kandidat/civilingenioer_i_robotteknologi/eksamen/afsluttendeopgaver/kandidat
- And in the master thesis course description:
  https://odin.sdu.dk/sitecore/index.php?a=fagbesk&id=83030 – 40 ECTS
  https://odin.sdu.dk/sitecore/index.php?a=fagbesk&id=83021 – 30 ECTS
  (These documents govern your thesis work, it would be unclever not to read the one relevant for you.)
- Some teachers provide additional material for their students which could also be of interest to others:
  - Poramate Manoonpong has an alternative Latex thesis template here: http://www.manoonpong.com/Thesistemplate/report.zip
  - Henrik Skov Midtiby provides some “scattered notes related to project work” here: https://source.coderefinery.org/henrikmidtiby/project-notes
  - Kjeld Jensen provides a “project survival guide” here: http://frobomind.org/studentprojects/doku.php